The $\frac{1}{3}\left(\frac{3}{4}\right)^k$ Distribution of Steiner Circuit Branch Lengths

Speculative Results Series, Paper 65

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Abstract

We define a *Steiner circuit branch* as a maximal sequence of consecutive Steiner circuits in a Collatz orbit, terminating at the first circuit whose exit value is $\equiv 5 \pmod 8$. Branches match the regular language $((7^*3)?1)^{k-1}(7^*3)?5$, where $k \geq 1$ is the total circuit count.

A naive analysis of the Steiner circuit state machine suggests that the 50/50 split at the 3-node should produce branch lengths geometrically distributed with parameter 1/2, giving $P(k) = (1/2)^k$. Empirical sampling of 10^5 branches via uniformly random entry points $n = 8t + 5$ instead yields the distribution:

$$P(\text{branch length} = k) \;=\; \frac{1}{3}\left(\frac{3}{4}\right)^k.$$

We interpret the observed coefficients 1/3 and 3/4 in terms of what they imply about the within-branch circuit entry distribution, report the empirically measured entry distribution, and state a conjecture that the distribution is exact. A first-principles proof is an open problem.

Status. This is a speculative results paper (stream 64+k). All empirical claims are supported by χ^2 goodness-of-fit tests. No new theorems are claimed. The proved foundations are in Paper 33 [1].

Contents

1 Background and Notation

We work with odd positive integers throughout. The *2-adic valuation* $v_2(n)$ is the largest power of 2 dividing n.

Definition 1.1 (Syracuse map)**.** The *Syracuse map* $S : \mathbb{O} \to \mathbb{O}$ is

$$S(n) \;=\; \frac{3n+1}{2^{v_2(3n+1)}},$$

which maps odd positive integers to odd positive integers by performing one odd Collatz step followed by all mandatory even halvings.

1.1 Mod-8 Step Taxonomy

Every odd integer n falls into one of four residue classes modulo 8, each corresponding to a distinct step type under S:

n mod 8	$v_2(3n+1)$	Step type	Exit class $S(n)$ mod 8
1	2	OEE	1 (always)
3	1	OE	1 or 5
5	≥ 3	OEEE$^+$	5 (always)
7	1	OE	3 (always, to the 3-node)

The key structural fact is that nodes 3 and 7 both route to the *3-node*: entry via 7 reaches 3 after a run of $v_2(n+1) - 2$ self-loop steps (Theorem proved in Paper 33 [1]), and entry via 3 is at the 3-node directly.

1.2 Steiner Circuits

Definition 1.2 (Steiner circuit)**.** Let $a_0 \in \mathbb{O}$ with $a_0 + 1 = 2^\alpha m$, m odd, $\alpha \geq 1$. The *Steiner circuit* starting at a_0 is the path of α consecutive Syracuse steps from a_0 to

$$b \;=\; \frac{3^\alpha m - 1}{2^\beta}, \qquad \beta = v_2(3^\alpha m - 1).$$

The *Steiner parameter* $\alpha = v_2(a_0 + 1)$ counts the odd steps; the *excess* $\beta = v_2(3^\alpha m - 1)$ counts the additional even steps in the final descent. The word of the circuit in the mod-8 alphabet is the sequence of residues $a_0 \bmod 8, a_1 \bmod 8, \ldots, a_{\alpha-1} \bmod 8$ (one symbol per odd step).

Remark 1.3 (Steps vs. circuits). The mod-8 taxonomy table (Section 1) describes individual Syracuse *steps*, whereas a Steiner circuit groups $\alpha = v_2(a_0+1)$ consecutive steps into a single unit. For example, if $a_0 = 3$ then $\alpha = v_2(4) = 2$, so the circuit starting at 3 consists of *two* steps: $3 \to S(3) = 5 \to S(5) = 1$, with exit class $1 \pmod 8$. The taxonomy table entry for $3 \pmod 8$ describes only the *first* of these steps; the Steiner circuit definition binds all α steps into a single unit with a single exit value b.

Proposition 1.4 (Exit class, proved in Paper 33). *For any Steiner circuit with entry a_0:*

1. *If $a_0 \equiv 1 \pmod 8$, then $b \equiv 1 \pmod 8$ (OEE, $\beta = 1$).*
2. *If $a_0 \equiv 5 \pmod 8$, then $b \equiv 5 \pmod 8$ (OEEE$^+$, $\beta \geq 2$).*
3. *If $a_0 \equiv 3$ or $7 \pmod 8$, then $b \equiv 1$ or $5 \pmod 8$ with the split determined by the 3-node (see Lemma 1.5 below).*

Lemma 1.5 (3-node split). *Let $n \equiv 3 \pmod 8$, write $n = 8a + 3$. Then:*

$$S(n) \;=\; 12a + 5.$$

This satisfies $12a + 5 \equiv 1 \pmod 8$ if and only if a is odd, which holds if and only if $n \equiv 11 \pmod{16}$. Exactly half the residues in the class $\{n : n \equiv 3 \pmod 8\}$ satisfy $n \equiv 11 \pmod{16}$, so the split between exit via 1 and exit via 5 is exactly $1/2$ *— structurally determined, not probabilistic.*

Proof. $S(n) = (3n + 1)/2^{v_2(3n+1)}$. For $n = 8a + 3$: $3n + 1 = 24a + 10 = 2(12a + 5)$. Since $12a + 5$ is odd, $v_2(3n + 1) = 1$, giving $S(n) = 12a + 5$. Now $12a + 5 \equiv 5 + 4a \pmod 8$, which is $\equiv 1 \pmod 8$ iff $4a \equiv 4 \pmod 8$ iff a is odd iff $n = 8a + 3 \equiv 11 \pmod{16}$. Among $\{3, 11\} = \{n \bmod 16 : n \equiv 3 \pmod 8\}$, exactly one satisfies $n \equiv 11 \pmod{16}$, confirming the 50/50 split. □

1.3 The Steiner Circuit State Machine

The mod-8 entry class of each circuit is the mod-8 exit class of the previous one. This defines a Markov-like transition on $\{1, 3, 5, 7\}$:

Entry	Visits 3-node?	Exit
1	No	1 (certain)
3	Yes	1 or 5 (1/2 each)
5	No	5 (certain)
7	Yes (via 7-run then 3)	1 or 5 (1/2 each)

Crucially, exits are *always* in $\{1, 5\}$. States 3 and 7 are therefore *transient*: they can appear as entry states only at the very start of an orbit (when the initial value is $\equiv 3$ or $7 \pmod 8$) and are never re-entered thereafter. After at most one circuit, the chain is absorbed into $\{1, 5\}$.

2 Branches and the Naive Prediction

Definition 2.1 (Steiner circuit branch)**.** A *branch* is a maximal sequence of consecutive Steiner circuits $C_1, C_2, \ldots, C_k$ such that $C_1, \ldots, C_{k-1}$ each exit via $1 \pmod 8$ and C_k exits via $5 \pmod 8$. The *branch length* is $k \geq 1$ (total circuit count, including the terminal 5-exit circuit). In the mod-8 word alphabet, a branch of length k matches the language

$$((7^*3)?1)^{k-1}\,(7^*3)?5.$$

A branch of length $k = 1$ consists of a single OEEE$^+$ circuit (exit via 5, no preceding 1-exits).

2.1 The Naive Prediction

From the 3-node lemma (Lemma 1.5), every circuit that visits the 3-node independently terminates the branch with probability exactly 1/2. A first instinct is therefore:

$$P(\text{branch length} = k) \stackrel{?}{=} \left(\frac{1}{2}\right)^k.$$

This would mean $k-1$ consecutive non-terminating 3-node visits followed by one terminating visit, each with probability 1/2.

Why this is wrong. The empirical data decisively rejects this prediction (Section 4). The precise reason is an open problem.

3 Sampling Methodology

To measure the branch length distribution without systematic bias from orbit structure or double-counting of shared tails, we use the following protocol:

1. Draw t uniformly at random from $[1, T_{\max}]$ with $T_{\max} = 10^{15}$.
2. Compute $n = 8t + 5$ (so $n \equiv 5 \pmod 8$).
3. Since $n \equiv 5 \pmod 8$, the Steiner circuit starting at n always exits via $5 \pmod 8$ (Proposition 1.4). This circuit is the terminal circuit of a branch, not the start of one. The *new branch* begins at $b_0 = S^{(\alpha)}(n)$, the odd exit value of the circuit starting at n.
4. Follow consecutive Steiner circuits from b_0 until the first exit via $5 \pmod 8$, counting the total number of circuits. Record this count as k.

We collected $N = 10^5$ independent samples. The key point is that n itself is *not* counted as part of the branch being measured — it serves only as a well-defined, freshly sampled anchor. This ensures: (a) independence between samples (no shared orbit tails), (b) $T_{\max} = 10^{15}$ is large enough that the distribution of n mod 16 within the $5 \pmod 8$ class is effectively uniform.

The entry state distribution of b_0 was measured empirically across 10^5 samples and found to be approximately uniform across all four odd residue classes: $1 \pmod 8$: 25.18%; $3 \pmod 8$: 25.02%; $5 \pmod 8$: 24.85%; $7 \pmod 8$: 24.94%. This confirms the sampling is unbiased at the branch start. The within-branch entry distribution is a separate measurement reported in Section 4.3.

4 Empirical Result

4.1 The Observed Distribution

k	Observed fraction	Predicted $(1/2)^k$	Fitted $\frac{1}{3}(3/4)^k$
1	0.24852	0.50000	0.25000
2	0.18930	0.25000	0.18750
3	0.14023	0.12500	0.14063
4	0.10402	0.06250	0.10547
5	0.07940	0.03125	0.07910
6	0.05935	0.01563	0.05933
7	0.04463	0.00781	0.04449
8	0.03413	0.00391	0.03337
9	0.02513	0.00195	0.02503
10	0.01864	0.00098	0.01877

Table 1: Empirical branch length distribution ($N = 10^5$ samples) vs. the naive $(1/2)^k$ prediction and the fitted $\frac{1}{3}(3/4)^k$ distribution.

The fitted distribution

$$\boxed{P(\text{branch length} = k) \;=\; \frac{1}{3}\left(\frac{3}{4}\right)^k}$$

is a valid probability distribution: $\frac{1}{3}\sum_{k=1}^{\infty}(3/4)^k = \frac{1}{3}\cdot 3 = 1$.

4.2 Statistical Tests

A χ^2 goodness-of-fit test (10 bins plus tail, 9 degrees of freedom) strongly rejects the naive $(1/2)^k$ prediction ($\chi^2 \approx 454300$, $p \approx 0$) and is fully consistent with the $\frac{1}{3}(3/4)^k$ distribution ($\chi^2 \approx 8$, $p > 0.4$).

4.3 Interpretation of the Coefficients

The decay rate 3/4. The fitted decay rate implies that each circuit within a branch independently terminates the branch with effective probability 1/4. Given the 3-node lemma (Lemma 1.5), the termination probabilities by entry class are:

- Entry via 1 (mod 8): exits via 1 with certainty; termination probability = 0.
- Entry via 3 or 7 (mod 8): visits the 3-node; terminates with probability 1/2.
- Entry via 5 (mod 8): exits via 5 with certainty; termination probability = 1. However, such circuits can only appear as the *terminal* circuit of a branch by definition, so their contribution to the within-branch entry distribution is structurally suppressed.

The effective termination probability 1/4 implied by the fitted 3/4 decay is therefore a constraint on the entry state distribution of non-terminal circuits within branches. What that distribution actually is, and why it produces exactly 1/4, is an open problem.

Empirically, the entry state distribution of circuits *within* branches (measured across 10^5 samples) is: entry via 1 (mod 8): $\approx 31.3\%$; entry via 3 (mod 8): $\approx 31.3\%$; entry via 5 (mod 8): $\approx 6.2\%$ (suppressed, as expected); entry via 7 (mod 8): $\approx 31.2\%$. The fraction visiting the 3-node (entries via 3 or 7) is $\approx 62.5\%$. Explaining why this distribution produces an effective termination probability of exactly 1/4 — consistent with the fitted 3/4 decay — is the open structural question.

The amplitude 1/3. For a geometric distribution with survival ratio $r = 3/4$ supported on $k \geq 1$:

$$\sum_{k=1}^{\infty} r^k = \frac{r}{1-r} = \frac{3/4}{1/4} = 3,$$

so the normalising prefactor is 1/3. This is a purely algebraic consequence of the decay rate; no additional assumption is required.

5 Open Problems

Conjecture 5.1 (Branch length distribution)**.** *For branches sampled by uniform random entry at $n = 8t + 5$, $t \in [1, T_{\max}]$ with $T_{\max} \to \infty$:*

$$P(\textit{branch length} = k) = \frac{1}{3}\left(\frac{3}{4}\right)^k, \qquad k = 1, 2, 3, \ldots$$

Conjecture 5.2 (Within-branch entry distribution)**.** *The entry state distribution of non-terminal circuits within branches — empirically observed to be approximately* 31% *each for* $\{1, 3, 7\}$ (mod 8) *and* 6% *for* 5 (mod 8) *— produces an effective termination probability of exactly* 1/4 *per circuit, consistent with the fitted* 3/4 *decay rate.*

Conjecture 5.1 follows from Conjecture 5.2 together with Lemma 1.5 and the independence of successive 3-node coin flips. A first-principles proof of either conjecture is an open problem.

Acknowledgements

This paper is part of the speculative results stream (64+k) of the Collatz programme. All claims are empirical observations awaiting first-principles proof. The companion rigorous stream (32+k) contains the proved foundations on which these observations rest. Paper 33 (architecture) provides the Steiner circuit decomposition and the 3-node lemma used here.

References

[1] J. Seymour, *A Regular Expression Language for the Collatz Graph* (Working Paper, Paper 33), 2026.

[2] J. Seymour, *A Toolkit for Collatz Path Analysis* (Paper 37), in preparation, 2026.